

CARPENTRAS, France

WMO: 075860

Lat: 44.0829N

Lon: 5.0592E

Elev: 344

StdP: 14.51

Time Zone: 1.00 (EUC)

Period: 99-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	22.6	25.5	12.2	10.4	29.1	17.1	13.2	32.4	21.7	43.7	19.7	45.4	2.9	110	0.448

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	27.5	96.9	70.2	93.9	69.5	91.1	68.6	72.9	90.8	71.5	88.7	70.1	86.6	7.0	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
67.5	102.2	78.4	66.1	97.4	77.4	64.7	92.5	75.9	36.6	91.0	35.3	88.5	34.2	86.9	79.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
18.9	16.6	14.6	DB	18.4	101.6	4.3	3.0	15.4	103.7	12.9	105.5	10.4	107.1	7.3	109.3	
			WB	17.6	75.9	4.4	1.4	14.4	76.9	11.9	77.6	9.4	78.4	6.2	79.4	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	58.3	42.2	43.8	50.6	56.7	63.8	71.7	75.9	74.6	67.4	59.6	49.7	43.0
	DBStd	13.22	6.33	6.39	5.67	5.13	5.25	5.30	4.61	4.83	5.01	6.12	6.79	6.58
	HDD50	828	251	184	60	7	0	0	0	0	0	7	87	232
	HDD65	3442	705	593	446	252	84	6	1	1	30	184	459	681
	CDD50	3873	11	11	79	208	427	650	803	763	523	305	78	15
	CDD65	1013	0	0	0	3	46	207	338	299	103	17	0	0
	CDH74	12196	0	0	8	119	683	2566	4106	3449	1130	135	0	0
	CDH80	5592	0	0	0	15	177	1169	2117	1732	369	13	0	0
	WSAvg	5.4	5.2	5.8	5.9	6.1	5.7	5.5	5.7	5.3	4.9	4.8	5.0	4.8
	PrecAvg	29.0	2.4	1.6	1.7	2.4	2.3	1.7	1.4	1.7	3.7	3.9	4.2	2.1
(15) Precipitation	PrecMax	43.8	6.9	4.4	4.2	4.8	5.7	3.9	3.8	4.6	10.3	6.9	10.2	6.9
	PrecMin	19.8	0.2	0.0	0.0	0.6	0.3	0.2	0.1	0.0	0.7	0.7	0.7	0.0
	PrecStd	6.8	1.7	1.2	1.2	1.4	1.4	1.0	1.0	1.1	2.0	1.8	2.8	1.7
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	62.8	66.7	74.9	82.1	89.3	98.4	99.7	101.2	92.2	81.6	69.3	62.6
		MCWB	52.9	52.6	55.5	61.5	66.8	71.2	71.6	71.3	68.7	64.6	58.9	54.4
	2%	DB	58.9	61.9	69.8	77.3	84.7	94.1	96.7	97.1	87.8	77.7	65.9	59.4
		MCWB	51.0	50.7	54.7	59.2	64.8	69.9	70.3	70.0	66.8	64.3	57.2	52.6
	5%	DB	56.2	58.6	66.6	73.9	81.1	90.8	94.1	93.2	84.3	74.4	63.5	56.8
		MCWB	49.2	49.1	53.2	58.1	63.2	68.6	69.5	69.6	65.7	62.7	56.1	50.8
	10%	DB	53.3	55.6	63.3	70.2	77.7	87.1	91.2	89.8	80.9	71.5	60.9	54.1
		MCWB	47.4	47.3	51.7	55.8	61.7	67.0	68.4	68.6	64.8	61.0	54.3	48.8
	0.4%	WB	54.4	53.9	58.0	63.3	69.0	74.8	75.0	74.1	71.2	67.7	61.0	55.8
		MCDB	60.6	63.1	68.5	77.7	84.1	95.3	95.1	92.9	84.4	77.5	66.1	61.0
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	51.9	51.9	56.0	60.8	66.6	71.8	72.9	72.5	69.4	65.6	58.6	53.5
		MCDB	57.3	59.8	67.1	75.1	80.7	89.5	91.3	90.1	82.0	74.7	64.0	58.1
	5%	WB	50.0	50.2	54.3	58.8	64.7	69.9	71.5	71.1	68.0	63.9	57.0	51.7
		MCDB	55.2	56.8	64.4	71.4	77.9	86.9	89.2	88.2	80.3	72.1	62.2	56.2
	10%	WB	47.9	48.2	52.6	56.8	62.8	68.2	70.0	69.7	66.3	62.1	55.4	49.4
		MCDB	52.7	54.3	61.7	67.9	75.1	84.1	86.7	86.5	77.6	69.6	60.0	53.4
(35) Mean Daily Temperature Range	5% DB	MDBR	17.8	20.2	22.8	23.4	23.9	26.3	27.5	27.5	24.5	20.9	17.8	17.1
		MCDBR	19.9	25.5	28.8	31.0	30.7	33.0	33.2	34.6	30.7	24.6	20.4	18.5
		MCWBR	13.0	15.6	15.8	15.6	13.8	12.9	12.3	13.0	12.8	12.8	12.8	12.3
		MCDBR	18.3	22.9	25.3	28.4	27.6	30.7	30.4	30.8	25.9	22.0	17.8	16.1
	5% WB	MCWBR	12.5	14.5	14.2	14.7	13.0	12.8	12.0	12.5	11.6	11.9	11.6	11.5
(40) Clear-Sky Solar Irradiance	taub		0.313	0.332	0.370	0.391	0.401	0.400	0.393	0.390	0.379	0.377	0.340	0.311
	taud		2.473	2.417	2.333	2.295	2.316	2.343	2.363	2.376	2.397	2.403	2.455	2.490
	Ebn at Noon		258	272	274	277	277	276	277	274	268	251	244	247
	Edh at Noon		23	28	35	39	40	39	38	36	33	29	23	20
(44) All-Sky Solar Radiation	RadAvg		534	833	1239	1604	1925	2247	2253	1936	1473	930	583	461
	RadStd		57	85	110	133	149	108	114	75	92	89	73	51

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (35 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page